Palak Jain [Website]

Senior Data Scientist | Computer Vision | Applied ML | Multimodal AI

New York, NY | jainpalak3286@gmail.com | **LinkedIn**

Senior Data Scientist with 6+ years of combined industry and research experience developing and productionizing machine learning systems across computer vision, NLP, and multimodal AI. Expertise in image forensics, deepfake detection, document intelligence and representation learning, with a track record of translating research into scalable production solutions.

Experience

Lead Data Scientist | Computer Vision & Applied AI — **February 2023 - Present**
Verisk Analytics — **Jersey City, NJ**

- **Deepfake & Image Forensics**
  - Designed and productionized a CNN-based deepfake detection system trained on synthetic images generated by DALL-E, Midjourney, and Stable Diffusion, achieving 80% precision with <2% false positives and enabling scalable fraud screening through a Dockerized inference application.
  - Developed a pixel-level image manipulation detection pipeline using deep CNNs and heatmap-based localization, achieving 94% accuracy with <1% false-positive rate and providing interpretable visual evidence for fraud investigators.
- **Internet Duplication & Similarity Detection**
  - Developed a hybrid image similarity pipeline combining ORB features, ResNet embeddings, and SSIM scoring to identify externally sourced claim images, achieving >90% precision and reducing manual review workload.
- **Automated Image Classification & Pre-Filtering System**
  - Led end-to-end development of a 32-class claims image classification system using CLIP embeddings and CNNs, automating irrelavant-image filtering and improving downstream fraud detection throughput by 30%.
- **ML Operations & Model Quality**
  - Owned dataset curation, error analysis, and iterative model improvement using FiftyOne, improving data quality and model performance across multiple fraud-detection pipelines.
  - Built Dockerized inference applications using Flask and Gradio to support scalable model demonstrations and real-time fraud-screening workflows.

Data Science Intern — **June 2022 - August 2022**
Verisk Analytics — **Jersey City, NJ**

- Built lightweight computer vision pipelines for face and text detection and PII redaction, optimizing inference with OpenVINO to achieve 85% recall at 20 images/second.
- Designed automated data-integration workflows that reduced the need to resurvey 20K+ underwriting cases, generating significant and operational cost savings.

Research Assistant — **January 2021 - December 2022**
Indiana University- Data Science — **Indianapolis, IN**

- Extracted causal relationships from 1M+ biomedical sentences using Semantic role labeling, dependency parsing, and statistical weighting.
- Developed a BiLSTM-attention model in PyTorch for causal relation extraction, achieving 0.98 ROC-AUC on benchmark datasets and outperforming prior baselines.
- Fine-tuned BERT, RoBERTa, SciBERT models on the CauseNet corpus, improving F1 score by 8% over baseline.
- Optimized transformer attention mechanisms to mitigate semantic drift, improving accuracy on time-sensitive biomedical facts.

Data Engineer — **January 2019 - November 2020**
Infosys Limited — **Hyderabad, India**

- Designed and optimized 350+ ETL workflows across 150+ tables, improving pipeline throughput by 200% through SQL optimization, partitioning, and parallelization laying the data foundation for analytics and ML initiatives.
- Catalogued financing data for e-contract utilization from OLTP datasets and unstructured data sources.

Patents & Publications

Patents

- **Systems and Methods for Detecting Artificial Intelligence Generated Images** - U.S. Patent No. US-20260024365-A1
  Named Inventor | Insurance Services Office | Granted 2026
- **Systems and methods for detecting, localizing, and visualizing manipulations of portable document format files** - U.S. Patent No. US-12694109-B2
  Named Inventor | Insurance Services Office | Granted 2026
- **Machine Learning Systems and Methods for Image Splicing Detection and Localization** - U.S. Patent No. US-2025004610
  Named Inventor | Insurance Services Office | Granted 2025

Publications

[1] Sydney Anuyah, Jack Vanschaik, Palak Jain, Sawyer Lehman, and Sunandan Chakraborty. ***An Empirical Study of Causal Relation Extraction Transfer: Design and Data***. 2025. URL: https://arxiv.org/abs/2503.06076.

[2] Jack VanSchaik, Palak Jain, Anushri Rajapuri, Biju Cheriyan, Thankam Thyvalikakath, et al. ***Using transfer learning-based causality extraction to mine latent factors for Sjögren's syndrome from biomedical literature***. 2023. URL: https://www.cell.com/heliyon/fulltext/S2405-8440(23)06473-3.

Technical Skills

Programming & Data	Python, Bash, SQL
Machine Learning	Supervised & Unsupervised learning, Model Evaluation, Hyperparameter Tuning, Error Analysis
Deep Learning	Pytorch, CNNs, RNNs, BiLSTM, Attention Mechanisms, Transformers
Computer Vision	OpenCV, Image Classification, Object Detection, Image Forensics, Heatmap Localization, Feature Matching OpenCV, OpenVINO Toolkit, FiftyOne
NLP & LLMs	Transformers (BERT, GPT etc.), Causal Inference, Semantic Search, Dependency Parsing, Embeddings, Fine-tuning
Multimodal & Applied AI	Vision-language models (CLIP), Multimodal Retrieval (text, image, video)
MLOPs & Deployment	Docker, Flask, Gradio, openVINO, FiftyOne
Cloud, Data & Analytics	AWS (EC2, S3), MySQL, SQL Server, ETL
Analytics & Statistics	Statistical Modeling, Hypothesis Testing, Predictive Modeling, Exploratory Data Analysis, PowerBI

Education

M.S. in Applied Data Science, Indiana University Indianapolis **2021 - 2022**
B.E. in Electronics and Telecommunication, Devi Ahilya University, India **2014 - 2018**